Ivar Cashin Eriksson

Curriculum Vitae

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Introduction

ML researcher and engineer with an MSc in Engineering Physics and seven years of experience. I want to build a research career developing trustworthy AI. My background combines deep learning, probabilistic modelling and numerical optimisation in radiotherapy planning with the development of reproducible ML systems. I am particularly interested in how foundation models can quantify uncertainty and support decisions in safety-critical settings such as power systems.

Education

2015–2020 **KTH, Royal Institute of Technology – MSc. in Engineering Physics**, GPA: 5.0 / 5.0
Specialisation: Applied Mathematics, Statistical Learning and Data Analytics.
Relevant coursework: Statistical Machine Learning, Reinforcement Learning, Optimisation.
Thesis: Probabilistic Dose Prediction for Radiation Therapy Planning

2019 **ETH Zürich, Swiss Federal Institute of Technology – Applied Mathematics**, *Exchange*

2017–2022 **SU, Stockholm University – BSc. in Economics**, *Additional Bachelor's Degree*

2017 **ZJU, Zhejiang University**, *Research exchange*, Laser trimming of microring resonators.

Selected Experience

Researcher within Data Science @ RaySearch Laboratories

- Developed autoencoder-based patient embeddings and learned similarity from segmented 3D CT scans.
- Predicted dose-volume histograms and spatial dose distributions using sparse Gaussian processes.
- Worked with medical physicists to define requirements and evaluate methods in safety-critical settings.

Keywords: probabilistic modelling, Gaussian processes, representation learning, multi-objective optimisation, clinical collaboration

Read more...

Data Scientist / ML Engineer @ Signum Life Science

- Develop automated ML workflows for pharmaceutical forecasting, from data validation and training to experiment tracking, batch inference, and reproducible evaluation.
- Design maintainable ML architecture supporting thousands of independently trained models.
- Develop embedding-based retrieval and matching workflows for domain-specific text, evaluating semantic similarity and analysing retrieval failures.

Keywords: Python, machine learning, data pipelines, model evaluation, Databricks, MLflow, Docker, Linux

Read more...

Principal Data Scientist @ Valcon

- Led ML projects across energy, manufacturing, etc., from problem formulation to deployment.
- Designed and deployed real-time neural net using transfer learning with text and categorical inputs.
- Mentored junior colleagues and delivered training in explainable AI, Git, and software architecture.

Keywords: transfer learning, explainable AI, production ML, technical mentoring, interdisciplinary collaboration

Read more...

Skills and Other

Technology	Python, PyTorch, TensorFlow, SQL, C++, Git, Linux, Docker, MLflow, Databricks
Methods	Probabilistic modelling, deep learning, representation learning, optimisation, forecasting
Language	Native English and Swedish, intermediate Danish

Detailed Work Experience

2024–Present **Data Scientist / ML Engineer**, *Signum Life Science*, Copenhagen

At Signum, I work across ML, software architecture, and project management, translating loosely defined problems in pharmaceutical analytics into maintainable production systems.

I develop embedding-based retrieval and matching workflows for domain-specific text. This involves evaluating semantic similarity, analysing retrieval failures, and improving matching reliability.

I also lead architecture and development of a forecasting platform for pharmaceutical prices, demand, and patient populations. The platform combines data validation, model training, experiment tracking, and batch inference in Databricks, supporting thousands of independently trained models. My work includes reproducible evaluation, maintainable software design, and coordination with technical and domain stakeholders.

2022–2024 **Principal Data Scientist**, *Valcon*, Copenhagen

At Valcon, I led ML projects across energy, IoT, pharmaceutical, and manufacturing domains, taking projects from problem formulation and model development through production deployment. I also mentored junior colleagues and delivered training in explainable AI, Git, and software architecture.

For one project, I designed and deployed an end-to-end pipeline for a real-time neural network. Using transfer learning, the model combined natural-language and high-dimensional categorical inputs to predict high-dimensional categorical outputs. I was responsible for the training pipeline, modular model architecture, evaluation, and production integration.

I received the Valcon People's Choice Award for exemplary work, leadership, team spirit, and scientific curiosity.

2021–2022 **Data Scientist**, *Valtech*, Copenhagen

At Valtech, I developed ML solutions for e-commerce, including time-series forecasting and customer-behaviour modelling. I built forecasting pipelines using Prophet and applied clustering to behavioural data to support personalisation and customer segmentation.

I worked with technical and commercial stakeholders to translate modelling results into operational recommendations and presented findings to senior decision-makers.

2019–2021 **Researcher within Data Science**, *RaySearch Laboratories*, Stockholm

At RaySearch Laboratories, I researched ML and optimisation methods for automated radiotherapy planning. The work combined deep learning, probabilistic modelling, and multi-objective optimisation in a safety-critical clinical setting.

I developed a pipeline that learned latent anatomical representations from segmented 3D CT scans using autoencoders. These representations were used to identify similar patients through a learned similarity metric and to predict dose-volume histograms and spatial dose distributions using sparse Gaussian processes.

I also redesigned lexicographic optimisation algorithms to better represent clinical decision hierarchies. I collaborated with medical physicists and clinicians to define requirements, evaluate model behaviour, and ensure that the resulting methods aligned with oncological standards and practical workflows.

Personal Projects

Project `iris` Developed a visual product retrieval system that detects products within webshop images and matches them to catalogue items. Combined YOLOv8 object detection with OpenCLIP embeddings and Qdrant vector similarity search. Built the supporting data collection pipeline, backend APIs, and Chromium extension to display product links within images. GitHub repository.